

Ventilator-Associated Pneumonia in the Neonatal Intensive Care Unit

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EDUCATION GAPS

Neonatal clinicians need to be aware of the increased morbidity and mortality caused by ventilator-associated pneumonia and be able to implement evidence-based strategies to prevent it.

OBJECTIVES *After completing this article, readers should be able to:*

1. Define ventilator-associated pneumonia in infants in the NICU.
2. Identify common pathogens causing ventilator-associated pneumonia.
3. Identify measures to prevent neonatal ventilator-associated pneumonia.

ABSTRACT

Intubated infants in the NICU are at risk of developing ventilator-associated pneumonia (VAP), a common type of health care–associated infection. The Centers for Disease Control and Prevention developed guidelines for diagnosing VAP in patients younger than 1 year, which include worsening gas exchange, radiographic findings, and at least 3 defined clinical signs of pneumonia. VAP in infants is treated with empiric antibiotics selected based on local resistance patterns and individualized patient data. Many NICUs have implemented prevention bundles in an effort to decrease VAP by ensuring the cleanest environment for intubated neonates (hand hygiene, sterile handling of equipment), positioning of infants to prevent gastric reflux, and constantly reevaluating for extubation readiness. Although these prevention bundle elements are intuitive and generally low risk, none are based on strong research support. This article reviews the epidemiology, pathogenesis, diagnosis, treatment, and prevention of VAP in NICU patients, focusing on recent evidence, highlighting areas of emerging research, and identifying persistent knowledge gaps.

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ABBREVIATIONS

CDC	Centers for Disease Control and Prevention
ETT	endotracheal tube
GER	gastroesophageal reflux
MAP	mean airway pressure
NHSN	National Healthcare Safety Network
VAE	ventilator-associated event
VAP	ventilator-associated pneumonia

INTRODUCTION

Ventilator-associated pneumonia (VAP) is a health care–associated infection that increases morbidity, mortality, and hospitalization costs. Infants in the NICU—especially those born prematurely and requiring prolonged hospitalization—are

vulnerable to VAP due to anatomic underdevelopment, repeated airway instrumentation, dysbiosis, and limited innate and adaptive immunity. Some epidemiologic studies suggest that VAP is the second most common health care–associated infection in the NICU after bloodstream infection. (1) However, the significant overlap of the clinical features of VAP with other pulmonary pathology common in infants, particularly preterm populations, can make a definitive diagnosis of VAP challenging. This article will review current knowledge about neonatal VAP and remaining uncertainties in the field. In this review, the term “neonatal VAP” encompasses VAP in any infant in the NICU, including those older than 30 days. Topics addressed include evolving clinical definitions, details of the pathophysiology and microbiology of neonatal VAP, and suggested best practices for prevention and treatment.

DEFINITION

VAP generally refers to bacterial (rather than viral or fungal) infection of the lungs in an intubated patient. As discussed herein, it can be difficult to diagnose neonatal VAP with certainty. The clinical, radiographic, and laboratory constellation of findings used to identify neonatal VAP are all nonspecific, and recent research suggests significant variability among clinicians evaluating possible cases. (2) This diagnostic uncertainty affects epidemiologic data and treatment plans, and can have reimbursement and medicolegal consequences, as VAP is increasingly seen as a preventable complication in older populations. (3)(4)

The most commonly used formal criteria for neonatal VAP derive from the Centers for Disease Control and Prevention (CDC) guidelines for VAP in adults and children, which do not consider neonatal-specific pathology such as respiratory distress syndrome or bronchopulmonary dysplasia (Table 1). (5) The CDC definition for VAP in children

younger than 1 year specifies mechanical ventilation for at least 2 days through an endotracheal tube (ETT) before a diagnosis of pneumonia is made.

Because of challenges inherent to diagnosing neonatal VAP, however, the National Healthcare Safety Network (NHSN) has transitioned away from the term, instead electing to monitor ventilator-associated events (VAE), which are more broadly defined. This change was adopted for adult populations in 2013, (6) then extended to pediatric (including neonatal) populations in 2021. (7) A pediatric VAE is defined by the NHSN as:

- Required:
 - Mechanical ventilation through an ETT with at least 2 calendar days of stable or decreasing daily minimum fraction of inspired oxygen (FiO_2) and mean airway pressure (MAP) values.
 - For patients younger than 30 days, MAP values of 0 to 8 cm H_2O are recorded as 8 cm H_2O for the purposes of surveillance; for patients 30 days of age or older, MAP values of 0 to 10 cm H_2O are recorded as 10 cm H_2O . This increases the stringency of VAE determination.
- Followed by at least 1 of the following:
 - Increased daily minimum FiO_2 at least 0.25 over the daily minimum FiO_2 recorded on the first day of the baseline period (for example, a patient who was requiring 0.21 FiO_2 would meet this criterion with an increase to an FiO_2 of 0.46).
 - Increased daily minimum MAP value at least 4 cm H_2O over the daily minimum MAP recorded on the first day of the baseline period.

It is important to note that the NHSN criteria for VAE were developed for population-wide surveillance. VAE is

Table 1. CDC Diagnostic Criteria for Neonatal VAP

Required
1. Worsening gas exchange (identified based on desaturations, need for increasing ventilator settings, and/or a rising FiO_2 requirement)
AND
2. Radiographic evidence of a new or worsening focal infiltrate, consolidation, cavitation, or pneumatocele
With at least 3 of these findings
• Temperature instability
• Leukopenia ($\text{WBC} < 4,000/\mu\text{L}$ [$< 4 \times 10^9/\text{L}$]) or leukocytosis ($\text{WBC} > 15,000/\mu\text{L}$ [$> 15 \times 10^9/\text{L}$]) with a left shift ($> 10\%$ band forms)
• Cough
• New onset of purulent sputum, change in character of sputum, or increased respiratory secretions requiring increased suctioning frequency
• Apnea or increased work of breathing (tachypnea, retractions, grunting, nasal flaring)
• Wheezing, rales, or rhonchi
• Bradycardia (< 100 beats/min) or tachycardia (> 170 beats/min)

CDC=Centers for Disease Control and Prevention, FiO_2 =fraction of inspired oxygen, VAP=ventilator-associated pneumonia, WBC=white blood cells.

not intended as a clinical diagnosis to guide management of specific patients. The NHSN decision to remove VAP as a publicly reported pediatric health outcome highlights the inherent challenges with the diagnosis, however. The agency's report on the new terminology refers to the subjectivity involved in radiographic and symptomatic diagnostic criteria for VAP. This subjectivity remains a challenge in the clinical setting. For this reason, the formal definition of neonatal VAP may continue to evolve as new diagnostic technologies become available.

Because research to date has been largely based on the CDC's definition of VAP, the remainder of this article adopts that framework and terminology.

EPIDEMIOLOGY

The reported incidence of VAP in neonates varies between 2.7 and 37.2 episodes per 1,000 ventilator days. (2) This wide range is likely due to the definitional challenges discussed earlier and differences in study populations. One carefully performed epidemiologic study that included bronchioalveolar lavage for identifying a microbiologic cause as part of the diagnostic criteria found that among 198 neonates intubated for more than 48 hours, VAP occurred at a rate of 10.9 per 1,000 ventilator days. (8)

VAP is more likely to occur in very-low-birthweight infants and in lower-resource medical settings. (9) Other high-risk patients are infants with bronchopulmonary dysplasia and those who require sedation (10) or reintubation. Blood transfusions have been shown to be a risk factor for VAP. A theorized mechanism is that transfusions may lead to VAP as a result of increased pulmonary edema. (11)(12) A retrospective cohort study examined children receiving ventilation, including patients in 5 NICUs in the United States, and compared those with VAP to those without. (12) This group found that NICU patients with VAP were more likely to have had a transfusion of any blood product, as well as to have undergone sedation, neuromuscular blockade, or a recent surgical procedure. As all these characteristics are comorbidities associated with prolonged, high-risk hospitalization, the independent contribution of any given risk factor to VAP is difficult to determine.

PATHOGENESIS

The oral and nasal cavities, throat, and pulmonary airways are not sterile spaces. During and after birth, they quickly become colonized by bacteria that constitute the microbiota of those anatomic niches. (13) The earliest colonizers tend to be dominated by *Staphylococcus* and *Ureaplasma*

species. (14) These founding bacterial populations are dynamic, shifting over time in response to environmental exposures, immune pressures, antimicrobial medications, and maturational factors that are still incompletely understood. (15) Eventually the profile of the airway microbiome settles into equilibrium, often characterizable as 1 of 6 community state types (16) dominated by either *Streptococcus* or *Staphylococcus* species. Regardless of the community state type, most bacteria in the microbiota of the upper respiratory tract are rarely pathogenic. However, mixed among these commensal species are potentially pathogenic bacteria. These express toxins, adhesins, antibiotic resistance genes, and immune evasion molecules that impart a tendency to invade and damage protected host compartments.

The lungs experience constant exposure to bacteria from the upper respiratory tract microbiome, which are inhaled and carried by gravity into the bronchi along with respiratory secretions. Physical, cellular, and chemical defenses ordinarily work in concert to clear the lower respiratory tract of microbes. However, the combination of anatomic immaturity, lung tissue injury, dysbiosis favoring pathogenic species, (17) and physical impediments imposed by an ETT can trap virulent bacteria in the lower airways, where they multiply, invade across tissue barriers, and trigger dangerous inflammatory cascades. This pathophysiology is the basis of VAP.

The organisms that most frequently cause neonatal VAP are *Pseudomonas aeruginosa*, *Enterobacter* species, *Klebsiella* species, and *Staphylococcus aureus*. (18)(19) Multiple antibiotic-resistant strains of these species are increasingly common, which can complicate effective treatment. (20) *Enterobacter* and *Klebsiella*, both of which are implicated in neonatal VAP, are enteric organisms that are uncommon constituents of a normal respiratory microbiota. An open area of ongoing research centers around exploring how these intestinal commensals migrate to the airways.

One theory holds that intubated neonates experience increased respiratory tract exposure to these and other enteric microbes from refluxed gastric fluids. There is modest, indirect evidence to support this hypothesis. Pepsin, a gastric enzyme, was found in tracheal aspirates of intubated premature infants, with higher levels detected among patients being fed. (21) This finding indicates gastric reflux penetrating the ETT.

However, other studies are less clear on whether direct transit of gastric bacteria into the airway represents a significant cause of VAP. In adults, a randomized controlled

study showed that prophylaxis with acid-suppressing medications like ranitidine or sucralfate decreased the rates of VAP. (22) Albert et al studied the effects of enteral nutrition and the use of acid-suppressing medications in a multicenter study of pediatric patients 30 days to 18 years old. (23) Interestingly, the patients receiving acid-suppressive medications had a higher incidence of VAP. This was thought to be due to changes in the gastric microbiota once these medications were started. Another single-center pediatric study found no change in incidence of VAP or the incidence of airway colonization with gram-negative bacilli after the use of acid suppressive medications. (24)

Parker et al reported a single-center trial on the use of gastric residual evaluations in the extremely preterm infant and their effects on nutrition and feeding-related complications. (25) In some NICUs, the decision to feed an infant is based on prefeed residual evaluations due to concern about feed intolerance, overfeeding, and aspiration if there are residual feedings in the infant's stomach. In this study, patients were randomized to a residual-based feeding strategy or no prefeed residual measurements; the authors did not find any change in the rates of VAP in the experimental group. (25)

A newer theory based on data emerging from large-scale, culture-free, longitudinal microbiota analyses proposes the existence of a "gut-lung axis" of bacterial colonization in the newborn. (26) Although still incomplete, this model suggests that evolution of the intestinal microbiome shapes global immune responses and inflammatory cascades in ways that can affect the composition of the lung microbiome. A corollary hypothesis is that intestinal dysbiosis due to immune abnormalities or significant antibiotic exposure may result in establishment of abnormal airway microbial populations, perhaps including increased abundance of bacteria typically found in the intestine. (27) This gut-lung microbiome axis remains an area of active study and its relation to neonatal VAP is uncertain.

The ETT itself is an important contributor to VAP pathogenesis, and VAP risk increases with each day an infant remains intubated. (3) The ETT bypasses the tongue, epiglottis, and vocal cords, which are important anatomic protections against microaspiration. In addition to serving as a conduit for bacteria from other sources, the tube acts as a reservoir for bacteria, which coalesce to form a persistent biofilm on the inner lumen and outer surface. (28) These biofilms can, in turn, seed the lower respiratory tract, whereas the tube impedes physiologic clearance of bacteria through coughing and ciliary action. (4) Respiratory culture specimens obtained from an ETT frequently grow

multiple organisms, which can reflect polymicrobial colonization, VAP caused by multiple organisms (ie, polymicrobial infection), or VAP caused by 1 organism out of a diverse colonizing population (see diagnosis section).

Limited production, persistence, and recycling of surfactant is the physiologic basis for respiratory distress syndrome, and several lines of evidence suggest that surfactant deficiency also promotes bacterial persistence and expansion in the neonatal lung. (29)(30)(31) Administration of surfactant to newborns with group B *Streptococcus pneumoniae* led to improved oxygenation with lower FiO₂ requirements. (29) In a randomized controlled trial examining the effects of surfactant replacement for adults with VAP, the group randomized to receive surfactant replacement had decreased pulmonary neutrophilic infiltrates—determined by bronchoalveolar lavage—compared to control patients, suggesting less damaging inflammation in the experimental group. In this small trial, no significant differences were found between the 2 groups in the duration of intubation or 30-day mortality rates. (31)

Although the patient's own enteric bacteria represent a likely cause of some neonatal VAP, the external hospital environment is another important reservoir of potentially infectious microbes. Evidence indicates that environmental surfaces and caregiver exposure can increase an infant's risk of developing VAP while intubated. Much of this evidence comes from observational reports of decreased neonatal VAP incidence after the adoption of prevention bundles, which are discussed in more detail in the prevention section. A major goal of VAP prevention bundles is to limit patient exposure to bacteria by ensuring consistent, effective hand hygiene, limiting opportunities for bacteria to enter the closed ventilator circuit by avoiding unnecessary equipment changes or other disconnection events, and decreasing standing water within the circuit to decrease potential reservoirs for bacterial expansion.

DIAGNOSIS

A presumptive diagnosis can be made using the CDC criteria described in the definition section. The most common clinical and radiographic findings among patients with VAP are (in order of decreasing frequency): radiologic consolidation (Fig 1), increased secretions, increased ventilator settings, radiologic infiltrate, increased oxygen needs, oxygen desaturations, and retractions. Thermal instability, tachypnea, apnea, and crackles on auscultation are rare presenting symptoms. (8)

Bacteremia can accompany neonatal VAP, and it is appropriate to obtain a blood culture specimen before

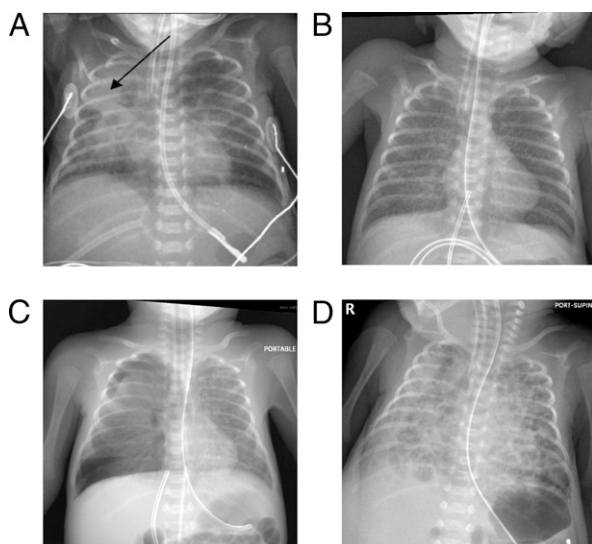


Figure 1. Neonatal ventilator-associated pneumonia (VAP) must be radiographically distinguished from other common radiographic findings. A lobar consolidation (A, arrow) or other radiographic evidence of neonatal VAP can coexist with or potentially be mistaken for noninfectious, common neonatal radiographic findings, such as respiratory distress syndrome (B), air leak (C), or bronchopulmonary dysplasia (D).

starting empiric antibiotic therapy for a patient with suspected pneumonia; however, a blood culture will not reliably identify a causative organism.

Because the respiratory tract is not a sterile site, a positive respiratory tract culture is not sufficient to diagnose infection. Polymicrobial colonization—including noninfectious colonization by any of the organisms listed before—is a normal finding, especially in patients who have undergone intubation for prolonged periods.

Bronchioalveolar lavage and bronchoscopic brush techniques have been used in older populations for more precise microbiologic identification of lower airway pathogens. (32)(33) Both techniques are less likely than upper airway sampling to yield contaminated, polymicrobial specimens, but both are impractical and carry potential procedural risks in low-weight neonates. As its name suggests, bronchoscopic brushing must occur under direct visualization, which typically cannot be performed through a small ETT. Bronchioalveolar lavage can be performed blindly, and some studies of hospitalized infants have used bacterial isolation from blind bronchioalveolar lavage to supplement the clinical, radiographic, and laboratory criteria of the CDC definition. (8) Although less complicated than bronchoscopy-assisted brush techniques and technically feasible through a small tube, blind bronchioalveolar lavage has not achieved the widespread availability necessary to include it as a diagnostic necessity.

Deep suctioned sputum samples—analyzed with Gram stain and culture with antimicrobial resistance testing on putative pathogens—can be modestly helpful but must be interpreted with care. The CDC describes purulent sputum as a sample that contains more than or equal to 25 neutrophils and less than or equal to 10 squamous epithelial cells per low-power field, (5) which is included in the diagnostic criteria for VAP. Purulent samples in which the Gram stain and culture are congruent and point to a known pathogen may be sufficient to establish a microbiologic diagnosis. In 1989, Slagle et al obtained serial tracheal aspirates from neonates who had prolonged intubation for more than 1 week. (34) The overwhelming majority (90%) of tracheal aspirates did not grow any bacteria for the first week. However, after the second week of intubation, they noted colonization by gram-positive bacteria, and after the fourth week by gram-negative organisms.

An effective diagnostic approach is to collect a sputum sample for analysis immediately before and 24 to 48 hours after initiation of empiric antibiotic therapy, and then compare the 2 samples for informative patterns. If the pretreatment sample is enriched by a potential pathogen that decreases after the initiation of antibiotics, and the patient improves clinically, that potential pathogen may have been the cause. If the pretreatment sample has a predominant organism that persists in the post-treatment sample, and the patient remains ill, it is important to consider whether that organism is effectively treated with the empiric antibiotic combination. Similar consideration must be given to the common scenario where the pretreatment sample contains a polymicrobial mix that shifts to a predominant (often antibiotic-resistant) organism after antibiotic initiation. In this case, one should again give careful thought to the post-treatment predominant species. If the patient remains ill, targeted therapy should be directed to that isolate.

Polymicrobial VAP is also possible. A NICU in Taiwan compared patients who were diagnosed with polymicrobial VAP to those in whom 1 organism was identified. (35) They reported that patients who suffered from polymicrobial VAP were more likely to be diagnosed with pneumonia later during their admission and tended to require longer durations of intubation. (35)

In patients with complicated NICU courses requiring prolonged intubation, it may be helpful to periodically survey sputum samples using Gram stain and culture in the absence of suspected infection. (36) This approach can provide baseline microbiota and antibiotic-resistance data, which can be useful when interpreting samples obtained later during VAP evaluations that might arise. A reasonable

approach is to gather 1 surveillance sample for each month of intubation. This is especially important for patients with histories of exposure to multiple antibiotics. Documentation of baseline colonization by resistant pathogens can help guide selection of initial empiric therapy of suspected VAP.

Several preliminary studies have explored the possibility of using chemical or cellular biomarkers to increase the sensitivity and specificity of VAP diagnosis. To our knowledge, only 1 clinical trial of a novel biomarker has been performed in an exclusively infant population to date. (37) This work examined the test characteristics of glutathione sulfonamide, which is a byproduct of neutrophil oxidative activity thought to correlate with infectious exposures. When levels of glutathione sulfonamide were assessed in culture-positive and culture-negative tracheal aspirates from intubated preterm infants, the culture-positive patients had fourfold higher measurements than the culture-negative controls. Glutathione sulfonamide levels were also shown to correlate well with 3-chlorotyrosine and myeloperoxidase levels, 2 other indicators of neutrophil-mediated oxidation. (37)

Other potential biomarkers for VAP include procalcitonin, (38)(39) several cytokines, (40)(41) and leukocyte activation markers such as soluble triggering receptor expressed on myeloid cells. (42)(43)(44) Biomarker measurement as a diagnostic test for neonatal VAP remains experimental, with insufficient high-quality evidence to support its clinical use.

MANAGEMENT

Suspected VAP should be treated with broad-spectrum empiric intravenous antibiotics. The antibiotic combination used for empiric treatment may be informed by local antibiogram data, the patient's colonization status and any history of past infectious isolates, or recent antibiotic courses that might have been selected for drug resistance. There is no gold standard empiric antibiotic prescription for suspected neonatal VAP. Two-drug combinations may be sufficient if they include broad coverage against gram-positive and gram-negative organisms, including drug-resistant strains common in nosocomial settings. Linezolid- or vancomycin plus- piperacillin/tazobactam or gentamicin will cover methicillin-resistant *S aureus* and a substantial portion of gram-negative organisms, though specific resistance to these combinations is also possible.

In severe cases, or if the patient is not responding as expected, a third agent may be added. Potential third-line antibiotics could include antipseudomonal cephalosporins (such as cefepime or ceftazidime), carbapenems, or

fluoroquinolones. If there is significant concern for aspiration pneumonia in a patient undergoing ventilation, antibiotics may be selected for high efficacy against anaerobes. Examples are ampicillin-sulbactam, clindamycin, piperacillin-tazobactam or meropenem. If polymicrobial infection is suspected based on suctioned sputum analyses, antibiotic therapy must be tailored to the suspected combination of pathogens. Early consultation with pediatric infectious disease subspecialists should be considered.

As with the selection of initial empiric antibiotics, there is no firm consensus on duration of treatment for neonatal VAP. Reported antibiotic courses range from 5 to 14 days. Recent evidence favors shorter durations. Chastre et al completed a randomized controlled trial on adults with VAP comparing antibiotic therapy for 8 days versus 15 days. (44) Both groups had similar rates of mortality and recurrent infections, and more importantly, patients who did develop recurrent infections were less likely to have multidrug-resistant organisms if they were in the shorter treatment group.

Cantey et al reported results of an extensive neonatal antibiotic stewardship program through which a "hard stop" for empiric antibiotics was introduced at 48 hours of treatment. (45) At that point, antibiotic continuation required reordering, and the recommended treatment duration for pneumonia (ventilator-associated or otherwise) was a total of 5 days. The group assessed antibiotic usage and outcomes for 2,502 patients: 1,607 during a baseline period before the stewardship program began and 895 during the postimplementation period. There was no significant change in the percentage of infants treated beyond 48 hours during the intervention period, but the proportion receiving 5 or fewer total days of antibiotics increased from 36% to 72%. Importantly, there was no corresponding increase in the number of patients treated for pneumonia requiring reinitiation of antibiotics within 14 days of the first course.

A recent publication by Goerens et al proposed a reasonable tripartite antibiotic stewardship program for infants with suspected VAP (Fig 2). (2) After 24 to 36 hours of empiric antibiotic coverage, this group recommends reevaluation of 1) clinical status, 2) laboratory values, and 3) microbiologic data from cultures and a tracheal aspirate Gram stain. If 1 or fewer domains suggests VAP, it is reasonable to consider ceasing empiric therapy and continuing clinical assessment. If 1 or 2 domains suggest VAP, continued therapy with appropriate antibiotics for an additional 48 to 72 hours may be considered, followed by reassessment. If all 3 domains are

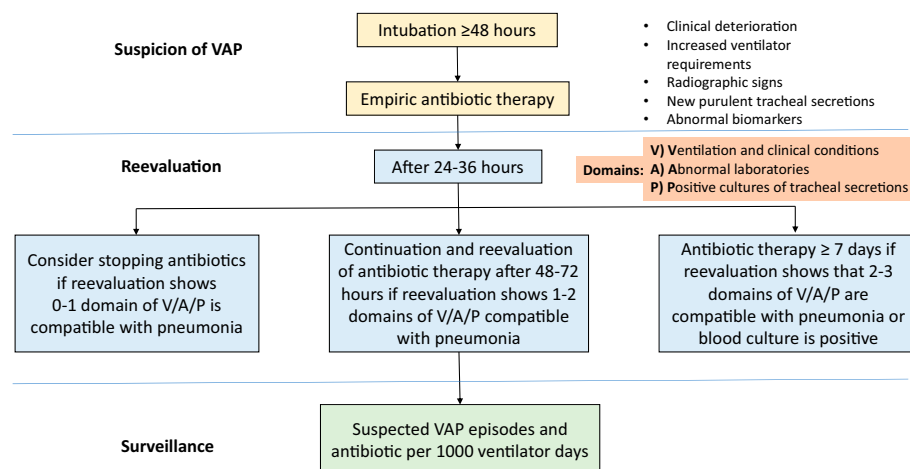


Figure 2. A proposed antibiotic stewardship protocol to rationally determine a patient's antibiotic course during a case of suspected ventilator-associated pneumonia (VAP). Decisions about whether to continue antibiotics are guided by clinical, laboratory, and microbiological reevaluation after a trial of empiric therapy. (Adapted from Goerens et al [2].)

indicative of VAP after a brief empiric trial, a full 7-day (or longer, depending on circumstances) course of antibiotics—narrowed, as feasible, to any microbial isolates—may be warranted.

PREVENTION

Multiple attempts have been made to prevent VAP using groups of preventive strategies as bundles. The Institute for Healthcare Improvement jumpstarted this initiative in 2004 when they started a campaign to protect patients from nosocomial infections, including VAP. They encouraged the use of evidence-based preventive measures including hand hygiene, avoiding intubation, decreasing length of intubation, head of bed elevation, and the use of cuffed ETTs in adults. (46)(47) Although neonatal VAP prevention bundle components are generally low-risk and intuitively sensible, there is little or no high-quality evidence from methodologically sound studies to support most VAP bundle components in isolation (Table 2). (48)

An observational study completed at a NICU in Egypt included criteria for intubation, improved hand hygiene practices, elevation of head of bed, and implementation of sterile handling of respiratory equipment including suction as preventive measures. (49) Introduction of the prevention bundle correlated with decreased rates of VAP, which fell from 67% of ventilated patients to 38% (36/1,000 ventilator days to 23/1,000 ventilator days). Although this is a significant decrease in rate, the baseline rates of VAP were quite high, which the authors attributed to overcrowding and understaffing (no respiratory therapists and high patient-to-nurse ratios). (49)

Another observational study in Spain used components similar to the previous bundle, however, they also included a section on feeding, in which prolonged feedings over 1 to 2 hours were preferred over bolus feedings or continuous feedings. (50) Bundle introduction correlated with a drop in VAP incidence from 11.7 per 1,000 ventilator days to 1.9 per 1,000 ventilator days. The bundle, however, did not decrease mortality or days of mechanical ventilation. Both studies used the CDC criteria for the diagnosis of VAP.

Jacobs Pepin et al recorded decreased VAP rates in a 60-bed NICU after the institution of a formal VAP bundle approach. (51) This group's bundle included "diligent hand hygiene, clean intubation technique, evaluation for extubation readiness, disinfection of environment and equipment, and management of bedside patient care routines." They recorded a decrease in VAP incidence from 8.5 to 2.5 cases per 1,000 ventilator days after the initiation of their quality improvement effort. (51)

A 2018 article by Gokce et al assessed VAP rates in a large tertiary care birth hospital in Turkey for 6 months before and 6 months after institution of a formal, 6-component neonatal VAP prevention bundle intended to limit environmental exposures that could increase infectious risk. Thirteen episodes of VAP were recorded (using the CDC definition for children <1 year of age) during the observation period. (52) Nine of the 13 events occurred during the prebundle observation period, representing a drop from 7.33 episodes per 1,000 ventilator days to 2.71 episodes per 1,000 ventilator days. The group also recorded adherence to elements of the VAP prevention bundle, which allowed statistical analysis to assess the

Table 2. VAP prevention bundle studies conducted with an exclusively neonatal population

Reference, year	Bundle Components	Type of Study	Results	Statistical Significance
Azab et al, 2015 (49)	Hand hygiene guidelines, oral hygiene, positioning, sterile suctioning and handling of respiratory equipment , strict guidelines on reintubation, change ventilator circuit if visibly soiled, daily evaluation of continued need for intubation and sedation	Prospective observational cohort Sample size 143	Decreased incidence from 36.3 to 23/1,000 vent days	$P=.0006$
Pinilla-González et al, 2021 (50)	Hand hygiene guidelines, oral care, positioning, provider training, sterile management of airway, tube feeding with breaks , disinfecting airway equipment, change ventilator circuit if visibly soiled, daily evaluation of continued need for intubation	Prospective observational cohort Sample size 280	Decreased incidence from 11.7 to 1.9/1,000 vent days	$P<.01$
Jacobs Pepin et al 2019 (51)	Hand hygiene guidelines, oral care, positioning, clean intubation technique, daily evaluation for need for intubation, guidelines on disinfection of patient environment, drain condensation from ventilator tube, guidelines on suctioning , change ventilator circuit when visible soiled.	QI	Decreased incidence from 8.5 to 2.5/1,000 vent days	$P<.004$
Gokce et al, 2018 (52)	Hand hygiene guidelines, oral care, positioning, daily evaluation of continued need for intubation, change ventilator circuit if visibly soiled.	Prospective observational cohort Sample size 431	Decreased incidence from 7.33 to 2.71/1,000 vent days	$P=.083$
Ceballos et al, 2013 (66)	Hand hygiene guidelines, oral care, positioning, replace entire circuit with reintubation, sterile field while intubating, frequent checks on position of tube, protocol to wean ventilator, separate oral and ETT suction catheters , clearing condensation in circuit.	QI	Decreased incidence from 8.9 to 3.9/1,000 vent days in patients <750 g Decreased incidence from 7.8 to 6.2/1,000 vent days in patients 751–1,000 g	NA
Tayel et al, 2017 (67)	Hand hygiene guidelines, daily evaluation of continued need for intubation and sedation, change ventilator circuit if visibly soiled, positioning, oral care. assessing for gastric residuals , clearing condensation in circuit.	Nonrandomized clinical trial Sample size 493	Decreased incidence from 24.6 to 19.9/1,000 vent days (based solely on bacteriological diagnosis of VAP)	NA

Bundle items that are unique to a study are highlighted in bold. ETT=endotracheal tube, NA=not available, QI=quality improvement, VAP=ventilator-associated pneumonia.

independent contribution of each bundle element on VAP rates. This approach showed that full adherence to all 6 components of the unit's prevention bundle was inversely correlated with VAP incidence, and that this relationship was statistically significant ($r=-0.04$, $P=.02$). The incidence of VAP was also inversely correlated with the number of components successfully implemented ($r=-0.07$, $P<.001$). Regression analysis of the individual components of the prevention bundle suggested that, of the approaches adopted, regular oral care with sterile saline was the only element with independent preventive significance. (52)

Many of the published VAP prevention bundles include head of bed elevation, presumably for reduction in gastroesophageal reflux (GER). This component derives from adult VAP bundles and is not an evidence-supported intervention to reduce GER in the neonatal population. The American Academy of Pediatrics report on the management of GER in preterm infants does not support the use of head elevation. (53) A small study of term infants suggested that head elevation actually increases the risk of GER compared to supine positioning. (54) For preterm infants, the only positioning interventions that have been shown to reduce GER are prone or left-side-down positioning. (55)

A classic study demonstrating the effect of environmental factors on nosocomial infection rates prospectively documented hospital-acquired pneumonia among 911 infants in the NICU at Boston Children's Hospital from 1974 through 1978. (56) Midway through the observation period, the unit moved from a small, crowded space to a new facility with 50% more staffing, fourfold more space per patient, and improved air filtration equipment. Coincident with this facility upgrade, the incidence of VAP dropped dramatically. Sixteen cases were recorded before the move, but only 1 afterward. (56) Although these observational studies cannot conclusively demonstrate causality, they are highly suggestive. Because careful hand hygiene, gentle oral care, and minimal handling of respiratory equipment are all low-risk, high-benefit components of quality care that improve patient safety, they should be universally adopted to reduce VAP rates and other potential complications.

ETT suction apparatuses that allow suctioning of the tube lumen without disconnecting the ventilator circuit (so-called "in-line" suction devices) were studied in 4 neonatal trials analyzed in a 2011 Cochrane meta-analysis. (57) No difference was found between in-line suction device use and "open" suctioning in terms of long-term mortality or morbidity. A prospective study by Cordero and colleagues (not included in the Cochrane review) reported a

nursing preference for the in-line system but demonstrated no significant difference in infectious or noninfectious complications between in-line and open suctioning among 175 randomly assigned preterm infants. (58)

Thorarinsdottir et al compared biofilms on 3 different types of ETTs, and reported reduced biofilm formation on silicone or noble metal-coated ETTs as compared to uncoated tubes. (59) Another study looked at removing biofilms using an antimicrobial photodynamic therapy on a patient model. (60) This group applied a photosensitive liquid that has antimicrobial properties through the ETT and then placed a fiberoptic catheter with a light source into the lumen of the ETT for a few minutes. This resulted in significant reduction in biofilm formation, though it is unclear how small the diameter of the fiberoptic catheter is and if it would fit into the small ETTs of premature neonates.

Studies have been conducted on oral health regimens and their relation to adult VAP, with some showing a protective effect. Many hospital-grade oral care products for adults contain antibacterial chemicals such as chlorhexidine, which are not recommended for infants. In neonates, several groups have examined the application of colostrum to the oral mucosa. The rationale is that innate antimicrobial and immunomodulatory molecules in colostrum will promote establishment and maintenance of a healthy airway microbiota. Whether this approach is effective remains uncertain due to conflicting results from similar studies. A systematic review published in 2020 analyzed 8 randomized trials assessing oral colostrum. (61) Three studies used a double-blinded, controlled design and 5 trials were single-blinded ($n=682$). Only 3 studies reported VAP as an outcome measure. Among these, there was a significant reduction in VAP (odds ratio 0.38, 95% confidence interval 0.17–0.88) in patients who received colostrum care. There was also a nonsignificant trend toward reduction in mortality, and the time to full enteral feedings was significantly shorter. In 2021, Aggarwal et al described a randomized clinical trial comparing oropharyngeal colostrum or donor breast milk with sterile water. Among 260 randomized infants, there were no significant differences in any primary or secondary outcome, including VAP, that depended on oral care regimen. (62)

Several common-sense practices included in some VAP prevention bundles include ensuring that at least 2 people are present during ETT retaping or repositioning; draining condensate from ventilator tubing every 2 to 4 hours; clearing secretions before ETT retaping, patient repositioning, or extubating; and rigorously ensuring that

laryngoscopes and newly placed ETTs are sterile and have not touched any environmental surfaces before use. As with all hospital-acquired infectious disease prevention, the single most important practice is rigorous, consistent hand hygiene before and after every patient contact.

OUTCOMES

Although VAP is undoubtedly a contributor to unfavorable neonatal outcomes, the exact degree of associated risk is hard to determine due to the difficulty of consistent diagnosis and overlap with other adverse clinical events.

Thattrimontrichai et al performed a prospective cohort study comparing outcomes and risk factors in neonates who developed VAP versus those who did not. (10) Of a group of infants admitted to a NICU in Thailand, 128 were included if they had been intubated for more than 48 hours. The group reported a VAP incidence of 13.3% (10.1 per 1,000 ventilator days) using the CDC definition for VAP in infants younger than 1 year. The patients diagnosed with VAP had statistically significant longer admission durations: 32 days compared to 14 days in patients who did not develop VAP, and therefore over double the incurred costs. This finding matches other studies indicating that neonatal VAP may prolong hospitalization. (63)(64)

Because of extensive overlap in populations, very little high-quality research has examined the relationship between neonatal VAP and development of bronchopulmonary dysplasia. Dell'Orto et al found higher rates of bronchopulmonary dysplasia in patients who had been diagnosed with VAP; however, when covariates were excluded from the statistical model the statistical significance of this difference was borderline ($P=.049$). (65)

CONCLUSION

Neonatal VAP is unquestionably a recognizable clinical entity that carries significant risk to the patient. However, basic unanswered questions about epidemiology, pathogenesis, and the impact of VAP on infant outcomes—beyond the common complications of prematurity and prolonged hospitalization—make VAP enigmatic and ripe for further study and characterization.

Fundamentally, VAP seems to result from introduction or trapping of pathologic bacterial subpopulations within a dynamic, evolving airway microbiota. Tissue injury, pooling secretions, microaspiration of gastric secretions, exposure to contaminated surfaces, dysbiosis due to pharmacologic or immunologic perturbations, and mechanical alterations

imposed by the endotracheal tube may all contribute to neonatal VAP. There is evidence to support each of these factors, yet not enough to establish a cohesive, unified model of how VAP arises.

Without such a model, determining optimal approaches to prevention is difficult. Bundle-based guidelines are generally safe and probably effective, in aggregate, in decreasing VAP burden, though the individual contribution of any particular bundle component is uncertain. An advantage of prevention bundles is that they are easier to implement as a group rather than as separate, disconnected practices; however, their bundling is, at some level, the very factor that prevents careful assessment of each element.

Treatment approaches for neonatal VAP also remain imprecise. Empiric antibiotic regimens must be evaluated in the context of the local antibiogram and the patient's own clinical history, along with microbiologic data from airway secretion samples, which must be interpreted with care. A movement to avoid prolonged antibiotic therapy for suspected neonatal VAP appears under way as part of a larger trend toward careful antibiotic stewardship. The trend toward limiting excessive antibiotic use, especially for conditions as nebulous as neonatal VAP, is perhaps the most important recent development in this field, given the recognized relationship between dysbiosis and poor outcomes.

Emerging research on the microbiome, immunomodulation, and biomarkers may expand and hone our understanding of neonatal VAP in the coming years. For now, it remains a disease that is challenging to effectively prevent, diagnose, and treat.

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American Board of Pediatrics Neonatal-Perinatal Content Specifications

- Know the effective techniques for control of health-care associated infection in the nursery, neonatal intensive care unit, and obstetrical unit.
- Know the pathogenesis and prevention of transmission of infections with multidrug-resistant bacteria
- Know strategies to prevent ventilator-associated pneumonia.

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1. There has been increased attention drawn to ventilator-associated pneumonia (VAP) in hospitalized patients. Neonates who undergo intubation with mechanical ventilation may be susceptible to ventilator-associated pneumonia (VAP). Which of the following statements concerning VAP in the NICU is correct?
 - A. The Centers for Disease Control and Prevention (CDC) have a specific consideration of neonatal-specific pathology, such as respiratory distress syndrome and bronchopulmonary dysplasia to qualify criteria for diagnosis of VAP.
 - B. The CDC definition for VAP in children younger than 1 year specifies mechanical ventilation for at least 2 days through an endotracheal tube before a diagnosis of pneumonia is made.
 - C. Oxygen requirement does not play a role in defining VAP for neonates.
 - D. As apnea is common in preterm infants, it is not considered in the criteria for diagnosis of VAP.
 - E. A positive blood culture coincident with worsening respiratory symptoms is required for identifying VAP.
2. VAP in neonates has been described in various populations, though reported rates from epidemiologic studies vary widely due to challenges in defining the condition in neonates. Which of the following is within the range of current incidence of VAP in neonates?
 - A. 1 episode per 1,000 ventilator days.
 - B. 2 episodes per 1,000 ventilator days.
 - C. 10 episodes per 1,000 ventilator days.
 - D. 50 episodes per 1,000 ventilator days.
 - E. 100 episodes per 1,000 ventilator days.
3. A preterm infant in the NICU has been receiving mechanical ventilation and has worsening symptoms suspected to be VAP. Which of the following organisms is a common cause of neonatal VAP?
 - A. *Pseudomonas aeruginosa*.
 - B. *Escherichia coli*.
 - C. *Mycobacterium tuberculosis*.
 - D. *Ureaplasma urealyticum*.
 - E. Group B *Streptococcus*.
4. It can be challenging to diagnose VAP using respiratory cultures, because positive growth may not necessarily indicate an infection. Using an approach of collecting a sputum sample for analysis immediately before and 24 to 48 hours after initiation of empiric antibiotic therapy with comparison of samples may be one strategy. Which of the following developments would increase the suspicion for VAP?
 - A. If the pretreatment sample has multiple organisms and the same organisms persist with no change in clinical status.
 - B. If the pretreatment sample has no growth and the post-treatment culture also has no growth.
 - C. If the post-treatment sample has no growth and the post-treatment culture yields a single organism that is reflective of skin flora.
 - D. If the pretreatment sample has organisms resistant to the antibiotics being used and the post-treatment sample no longer yields that organism.
 - E. If the pretreatment sample is enriched by a potential pathogen that decreases after the initiation of antibiotics and the patient improves clinically.

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5. A patient born at 25 weeks has severe bronchopulmonary dysplasia and is receiving mechanical ventilation at 12 weeks of age. With worsening symptoms and evaluation including laboratory and imaging, a diagnosis of VAP is considered. Which of the following is appropriate management for this patient?
- A. Close observation with no change in management.
 - B. Providing 100% fraction of inspired oxygen to increase lung function.
 - C. Broad-spectrum empiric intravenous antibiotics.
 - D. Intravenous acyclovir for 14 days.
 - E. Inhaled nitric oxide for 48 hours followed by sildenafil for 2 weeks.